

5% for adaptability—a trade-off that captures most efficiency gains without rendering the system unusable. While 17,000 tokens/sec is headline-grabbing, deeper analysis suggests:

- Performance gains come primarily from eliminating memory bottlenecks, not dramatically faster arithmetic
- Effective throughput depends heavily on:
 - Quantisation level (e.g., 4-bit vs 8-bit)
 - Batch size assumptions
 - Sequence length and KV cache reuse

In other words, HC1's advantage is structural, not just computational. It removes the dominant bottleneck rather than marginally improving compute units.

System-level advantages

At the system level, HC1 introduces several second-order benefits that are often underestimated:

- No HBM→no memory bandwidth bottleneck
- No large interconnect fabric→reduced latency variance
- No liquid cooling→dramatically simpler deployment
- Linear scaling with cards→predictable cluster performance

This leads to a key economic shift: AI infrastructure moves from bandwidth-constrained scaling to compute-density scaling.

Engineering trade-offs and hidden costs

While the HC1 architecture is highly innovative, it introduces significant constraints:

1. Model immutability. Updating weights requires new silicon. This creates:

- Long iteration cycles (months vs days)
- Tight coupling between the model lifecycle and fabrication

2. Manufacturing complexity. Mask-ROM-based designs:

- Increase tape-out rigidity

HC1 VS TRADITIONAL HARDWARE		
Feature	Traditional GPU/Accelerator	HC1
Model location	External memory	Hardwired in silicon
Data movement	High	Eliminated
Energy usage	High (memory dominated)	Low
Cooling	Liquid/advanced	Air-cooled
Flexibility	High	Limited
Throughput	<2000 tokens/sec	~17,000 tokens/sec

- Require near-perfect model finalisation pre-fabrication
- Introduce yield risks if defects affect encoded weights

3. Toolchain reinvention. HC1 effectively creates a new stack:

- Model→silicon compilers
- Hardware-aware quantisation pipelines
- New debugging paradigms (you can't 'print logs' from silicon)

4. Precision constraints. Most such systems rely on low-bit quantisation (≈4-bit), which:

- Works well for many LLMs
- But it may degrade accuracy in edge cases

For engineers, HC1 represents a deeper paradigm shift:

- Compute-in-memory becomes practical, not theoretical
- Software-hardware co-design becomes mandatory
- ASIC workflows evolve into model compilation pipelines

The biggest conceptual change: You no longer deploy models onto hardware; you fabricate hardware from models.

The HC1 architecture lends itself to:

- Data centres: High-throughput inference with lower energy footprint
- Enterprise AI: Predictable, cost-efficient deployments
- Edge AI (future): Real-time inference with minimal power budgets

Market implications and investment momentum

HC1 arrives at a pivotal moment when AI inference demand is rapidly accelerating, GPU supply

remains constrained, and energy consumption is emerging as a primary limiting factor in scaling AI systems. Its positioning as a cost-per-token disruptor is rooted in a fundamentally different

approach: eliminating the energy overhead of memory movement, removing dependence on high-bandwidth memory, and simplifying overall system architecture. These advantages not only improve efficiency but also catalyse broader industry interest in model-specific silicon, compute-in-memory designs, and domain-specific AI accelerators.

More importantly, HC1 does not represent an incremental step forward; it inverts the traditional compute paradigm. Instead of running models on chips, it embodies a model-as-chip architecture in which software is effectively integrated into hardware. This shift trades flexibility for substantial efficiency gains and replaces bandwidth-driven scaling with structural optimisation at the silicon level. If this approach proves scalable, it has the potential to disrupt a landscape long dominated by GPUs and general-purpose accelerators, offering a fundamentally new blueprint for computation.

In this sense, HC1 signals a decisive transition from programmable, general-purpose computing towards fixed, purpose-built intelligence hardware. For engineers, the implication is profound: the boundary between model and machine is rapidly dissolving, and future chip design may increasingly begin not with transistors, but with neural network architectures themselves. **EFY**

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